Forward

Forward retiming pulls registers toward outputs , decrease r(v) on the chosen node

The idea

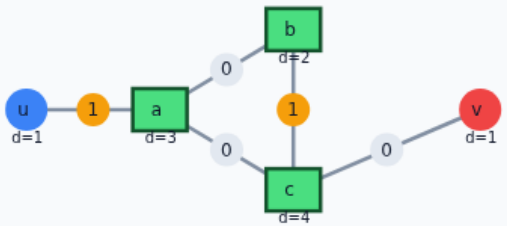
- Forward_retime('a', r, 1) sets r(a)=−1; e0 w_r becomes 0; e1/e4 gain registers

Forward −r

FORWARD RETIME

Forward −r

Step 1 / 5 · def · module02-03-forward-retime



Explanation

Forward retiming decreases r(v): registers move toward outputs across v.

- −r(v)

```
toward PO
```

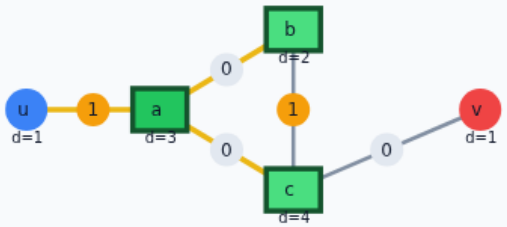
Forward −r

Pick a

FORWARD RETIME

Pick a

Step 2 / 5 · pick · module02-03-forward-retime



Explanation

Select node a — the first comb after the PI register.

- node a

```
d=3
```

Pick a

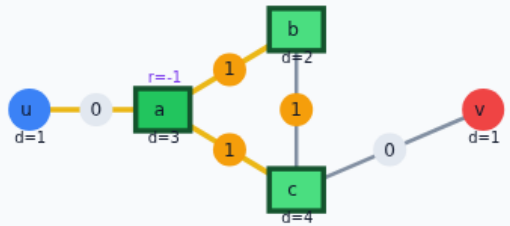
Apply forward

FORWARD RETIME

Apply forward

Step 3 / 5 · apply · module02-03-forward-retime

P=5 regs=3



Explanation

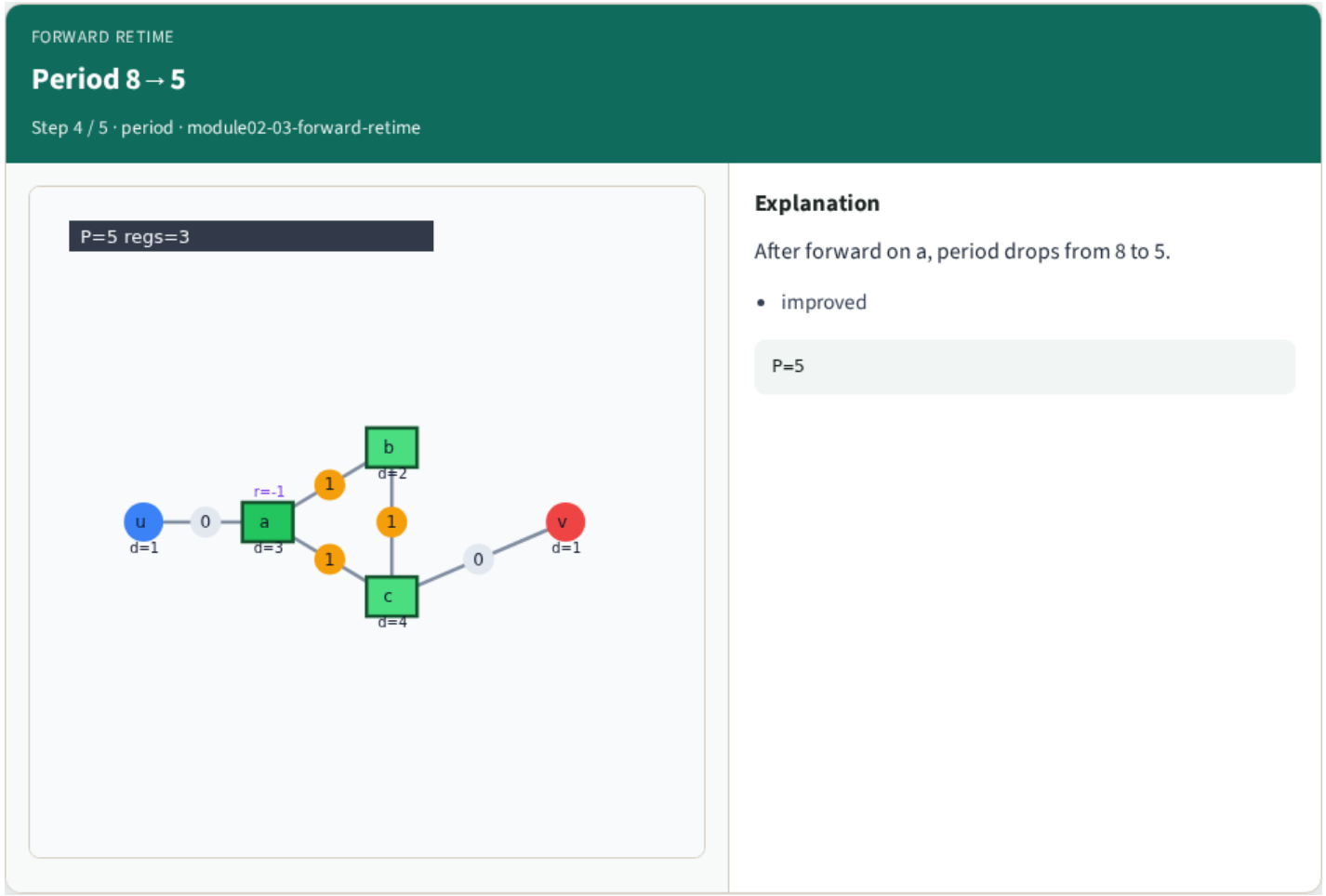
Set r(a)=−1. Edge weights become w + r(dst) − r(src).

- w_r

```
formula
```

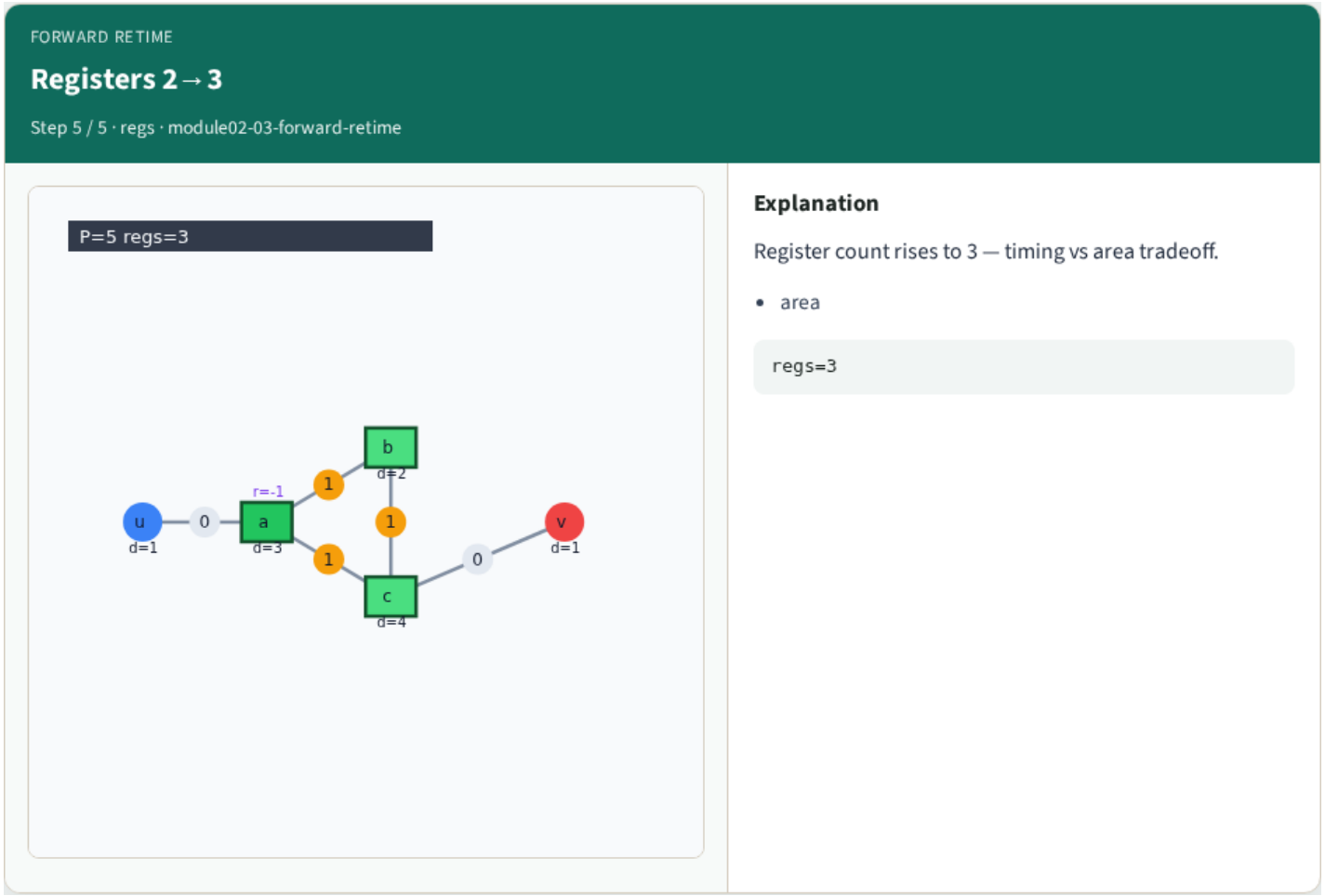
Apply forward

Period 8→5



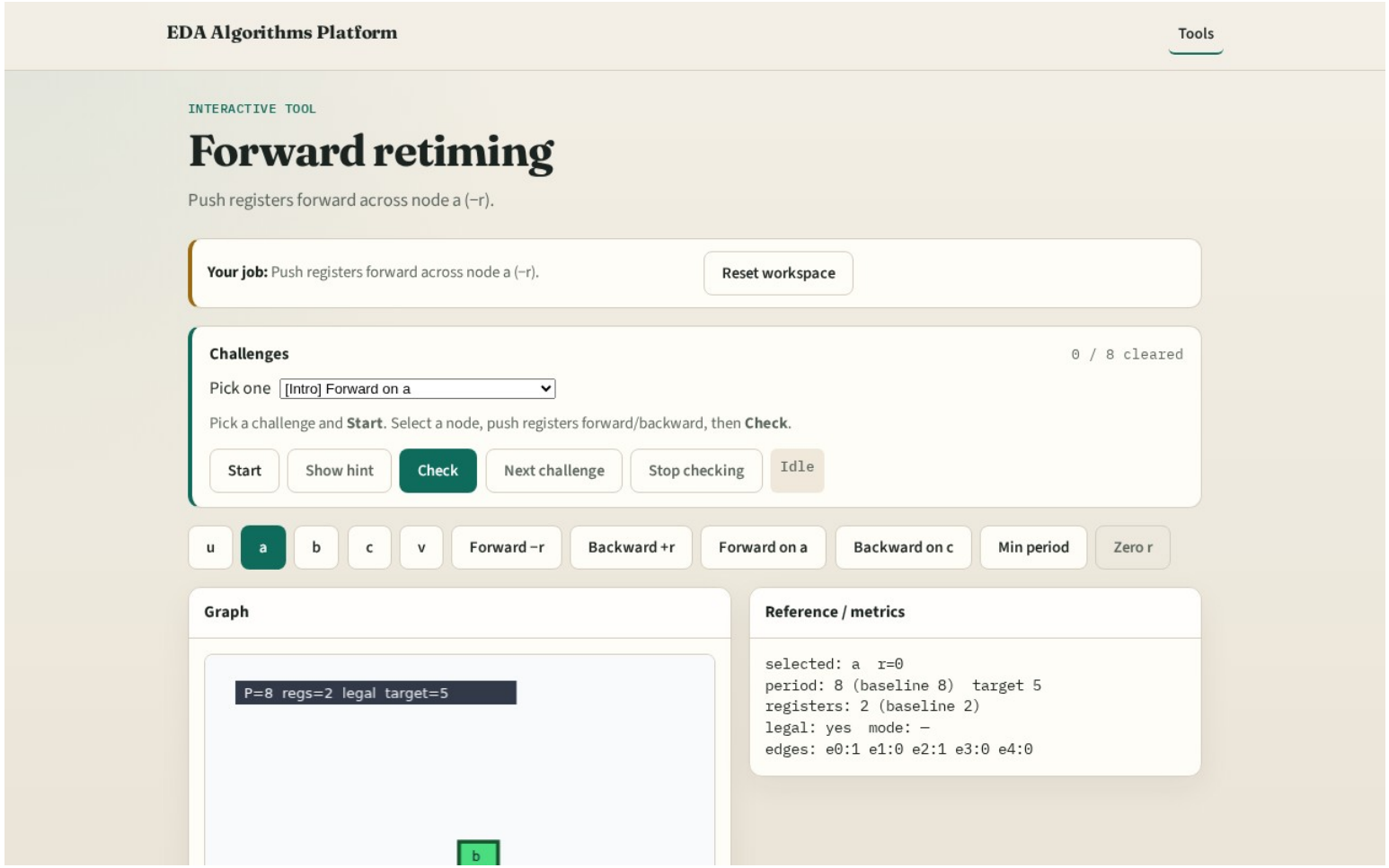
Period 8→5

Registers 2→3



Registers 2→3

Browser lab track



Browser lab starter

Implement track

- Call forward_retime('a', r, 1); print apply_retime edges and clock_period

Pitfalls

- Decreasing r on PI/PO
- Applying forward on c first and breaking legality

Your turn

- Next: backward retiming on c